

Assignment 5

Author: Jing Zhang

2/7/2017

Preliminaries

Package Loading

```
library(purrr)
```

```
## Warning: package 'purrr' was built under R version 3.2.5
```

```
library(dplyr, warn.conflicts = F)
```

```
## Warning: package 'dplyr' was built under R version 3.2.5
```

```
library(devtools)
```

```
## Warning: package 'devtools' was built under R version 3.2.5
```

```
install_github('jennybc/repurrrsive',force=TRUE)
```

```
## Downloading GitHub repo jennybc/repurrrsive@master
```

```
## from URL https://api.github.com/repos/jennybc/repurrrsive/zipball/master
```

```
## Installing repurrrsive
```

```
## '/Library/Frameworks/R.framework/Resources/bin/R' --no-site-file \
```

```
## --no-environ --no-save --no-restore --quiet CMD INSTALL \
```

```
## '/private/var/folders/_m/_pytmsvd1t16mbk79wkx275r0000gn/T/Rtmp0MD2uV/devtools335b761f2fde/jennybc-
```

```
## --library='/Library/Frameworks/R.framework/Versions/3.2/Resources/library' \
```

```
## --install-tests
```

```
##
```

```
library(repurrrsive)
```

```
library(broom)
```

```
## Warning: package 'broom' was built under R version 3.2.5
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 3.2.5
```

Assignment Part I

1) *Answer to Problem 1* (14.1.2)

$$\begin{aligned}\frac{\partial f}{\partial x} &= \ln(1+y^2) \\ \frac{\partial f}{\partial y} &= \frac{2xy}{1+y^2} \\ \frac{\partial^2 f}{\partial x^2} &= 0 \\ \frac{\partial^2 f}{\partial y^2} &= \frac{2x(1-y^2)}{(1+y^2)^2} \\ \frac{\partial}{\partial x} \frac{\partial f}{\partial y} &= \frac{2y}{1+y^2} \\ \frac{\partial}{\partial y} \frac{\partial f}{\partial x} &= \frac{2y}{1+y^2}\end{aligned}$$

Since f is a smooth function the mixed derivative theorem applies. At each point (x, y) in the plane we may define the gradient vector

$$Df(x, y) = \begin{bmatrix} \ln(1+y^2) \\ \frac{2xy}{1+y^2} \end{bmatrix}$$

The Hessian Matrix is

$$D^2f(x, y) = \begin{bmatrix} 0 & \frac{2y}{1+y^2} \\ \frac{2y}{1+y^2} & \frac{2x(1-y^2)}{(1+y^2)^2} \end{bmatrix}$$

In particular, setting $x = 1$ and $y = -2$ gives

$$Df(x, y) = \begin{bmatrix} \ln(5) \\ -0.8 \end{bmatrix}$$

and

$$D^2f(x, y) = \begin{bmatrix} 0 & -0.8 \\ -0.8 & -0.24 \end{bmatrix}$$

2) *Answer to Problem 2* (14.2.2)

$$\begin{aligned}\frac{dz}{dt} &= \frac{dz}{dx} \frac{dx}{dt} + \frac{dz}{dy} \frac{dy}{dt} \\ &= 6(3x^2y^4 + e^y) - 3(4x^3y^3 + xe^y)\end{aligned}$$

3) *Answer to Problem 3* (16.1.5)

The gradient vector is

$$Df(x, y) = \begin{bmatrix} 6x + 12y \\ -3y^2 + 12x - 36 \end{bmatrix}$$

The Hessian Matrix is

$$D^2f(x, y) = \begin{bmatrix} 0 & 12 \\ 12 & -6y \end{bmatrix}$$

When $Df(x, y) = 0$, we have

$$\begin{aligned}x + 2y &= 0 \\ y^2 - 4x + 12 &= 0\end{aligned}$$

Thus, we get

$$\begin{aligned}x = 4, y = -2, f(x, y) &= 32 \\ x = 12, y = -6, f(x, y) &= 0\end{aligned}$$

$(4, -2, 32)$ and $(12, -6, 0)$ are two critical points

We determine the nature of critical points by evaluating D^2f at each point.

$$D^2f(4, -2) = \begin{bmatrix} 6 & 12 \\ 12 & 12 \end{bmatrix}$$

has positive diagonals, and its determinant is $6 \times 12 - 12 \times 12 < 0$, therefore $(4, -2, 32)$ is a saddle point.

$$D^2f(12, -6) = \begin{bmatrix} 6 & 12 \\ 12 & 36 \end{bmatrix}$$

has positive diagonals, and its determinant is $6 \times 36 - 12 \times 12 > 0$, therefore $(12, -6, 0)$ is a local minimum point.

4) **Answer to Problem 4** (16.2.6)

The sum of three convex functions is convex. e^x , e^y , e^z are all convex functions.

5) **Answer to Problem 5** (17.2.1) In the case,

$$\begin{aligned}f(x, y, z) &= 2x + 3y + z \\ g(x, y, z) &= x^2 + y^2 + z^2 - 9\end{aligned}$$

The Lagrangian is

$$\begin{aligned}L(x, y, z, \lambda) &= f(x, y, z) - \lambda g(x, y, z) \\ &= (2x - \lambda x^2) + (3y - \lambda y^2) + (z - \lambda z^2) - 9\lambda\end{aligned}$$

At the critical point of the Lagrangian, we have Substituting into the constraint, we see that

$$x = \frac{1}{\lambda}; y = \frac{3}{2\lambda}; z = \frac{1}{2\lambda}$$

Substituting into the constraint, we see that

$$x = \pm \frac{3\sqrt{14}}{7}; y = \pm \frac{9\sqrt{14}}{14}; z = \pm \frac{3\sqrt{14}}{14}$$

The constrained minimum value of the function we have is $-3\sqrt{14}$. The constrained maximum value of the function we have is $3\sqrt{14}$.

Assignment Part II

6) *Answer to Problem 6*

```
PersonnelData <-data.frame(Representative =c(1:4), Sales=c(95,110,115,90), Territory=c(1:4))
PersonnelData
```

```
##   Representative Sales Territory
## 1             1    95          1
## 2             2   110          2
## 3             3   115          3
## 4             4    90          4
```

```
# mapply stands for 'multivariate' apply. Its purpose is to be able to vectorize arguments to a function
# that is not usually accepting vectors as arguments. In short, mapply applies a Function to Multiple L
mapply(class,PersonnelData[,1:3])
```

```
## Representative      Sales      Territory
##      "integer"      "numeric"      "integer"
```

7) *Answer to Problem 7*

In regression 1, x is the independent variable and y is the dependent variable; in regression 2, y is the independent variable while x is the dependent variable. Since the regressors are different, the two regressions differ.

```
x<-1:10
y<-1/3*x+rnorm(10)
lm(y~x-1)
```

```
##
## Call:
## lm(formula = y ~ x - 1)
##
## Coefficients:
##      x
## 0.3345
```

```
lm(x~y-1)
```

```
##
## Call:
## lm(formula = x ~ y - 1)
##
## Coefficients:
##      y
## 2.624
```

8) *Answer to Problem 8*

How to get the coefficient of a multiple linear regressions? The formula is: the formula is: $b = (X'X)^{-1}(X')Y$

```
xbeta <-solve(t(x)%*% x) %*% (t(x)%*%y)
xbeta
```

```
##           [,1]
## [1,] 0.3345258
```

```
ybeta <-solve(t(y)%*%y) %*% (t(y)%*%x)
ybeta
```

```
##           [,1]
## [1,] 2.623846
```

```
# given a matrix or data.frame x, t returns the transpose of x
```

9) *Answer to Problem 9*

```
install.packages("fueleconomy",repos="http://cran.rstudio.com/")
```

```
##
## The downloaded binary packages are in
## /var/folders/_m/_pytmsvd1t16mbk79wkx275r0000gn/T//Rtmp0MD2uV/downloaded_packages
```

```
library(fueleconomy)
```

```
lm(vehicles$cyl ~ 1)
```

```
##
## Call:
## lm(formula = vehicles$cyl ~ 1)
##
## Coefficients:
## (Intercept)
##          5.772
```

```
mean(vehicles$cyl,na.rm=T)
```

```
## [1] 5.771867
```

10) *Answer to Problem 10*

```
lm(cyl~fuel-1,data=vehicles)
```

```
##
## Call:
## lm(formula = cyl ~ fuel - 1, data = vehicles)
##
## Coefficients:
##           fuelCNG           fuelDiesel
```

```
##           6.517           6.365
##           fuelGasoline or E85      fuelGasoline or natural gas
##           7.083           6.889
##           fuelGasoline or propane      fuelMidgrade
##           8.000           7.953
##           fuelPremium      fuelPremium and Electricity
##           6.363           6.000
## fuelPremium Gas or Electricity      fuelPremium or E85
##           4.571           7.500
##           fuelRegular      fuelRegular Gas and Electricity
##           5.450           4.000
```

```
avgcylbyfuel <- vehicles %>% na.omit() %>% group_by(fuel) %>% summarise(meancyl=mean(cyl))
avgcylbyfuel
```

```
## # A tibble: 12 × 2
##           fuel meancyl
##           <chr>   <dbl>
## 1           CNG 6.517241
## 2          Diesel 6.364989
## 3 Gasoline or E85 7.083413
## 4 Gasoline or natural gas 6.888889
## 5 Gasoline or propane 8.000000
## 6          Midgrade 7.953488
## 7          Premium 6.362771
## 8 Premium and Electricity 6.000000
## 9 Premium Gas or Electricity 4.571429
## 10 Premium or E85 7.500000
## 11          Regular 5.449887
## 12 Regular Gas and Electricity 4.000000
```

11) Answer to Problem 11

```
vehicles2 <- vehicles%>%mutate(newclass = ifelse(class == "Special Purpose Vehicle"| class == "Special Purpose Vehicle", "Special Purpose Vehicle", class))

vehicles2$year <- vehicles2$year - min(vehicles2$year)

splitclass <- vehicles2 %>% split(.$newclass)

m <- cbind(as.data.frame(
  splitclass %>%
    map( ~ lm(hwy ~ year, data=.x)) %>%
    map(summary) %>%
    map(c("coefficients")) %>%
    map_dbl(2)),
  as.data.frame(
    splitclass %>%
      map( ~ lm(hwy ~ year, data=.x)) %>%
      map(summary) %>%
      map(c("coefficients")) %>%
      map_dbl(4)))

names(m) <- c("coefs", "std.err")
```

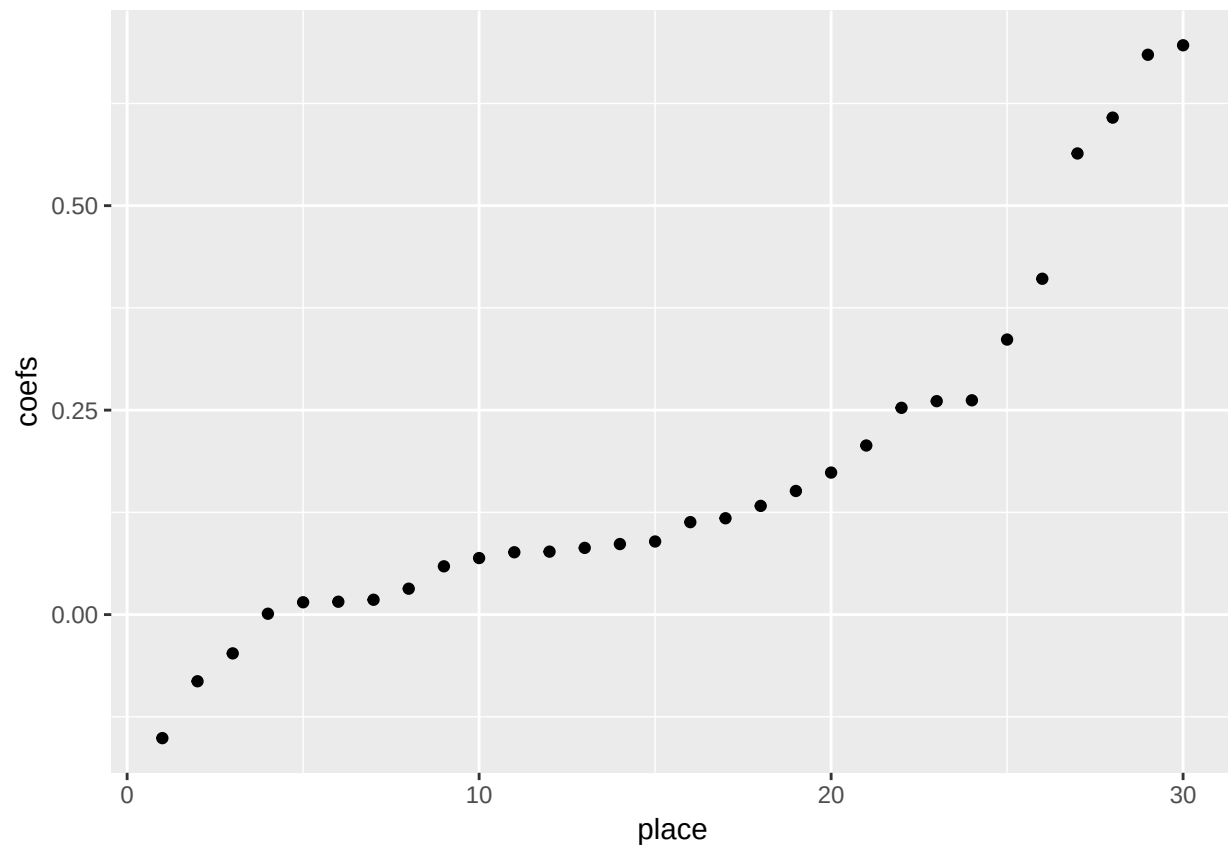
```

m <- m[order(m$coefs), ]

m$place <- seq(1:length(m$coefs))

#plotting
ggplot() +
  geom_point(data=m, mapping=aes(x=place, y=coefs))

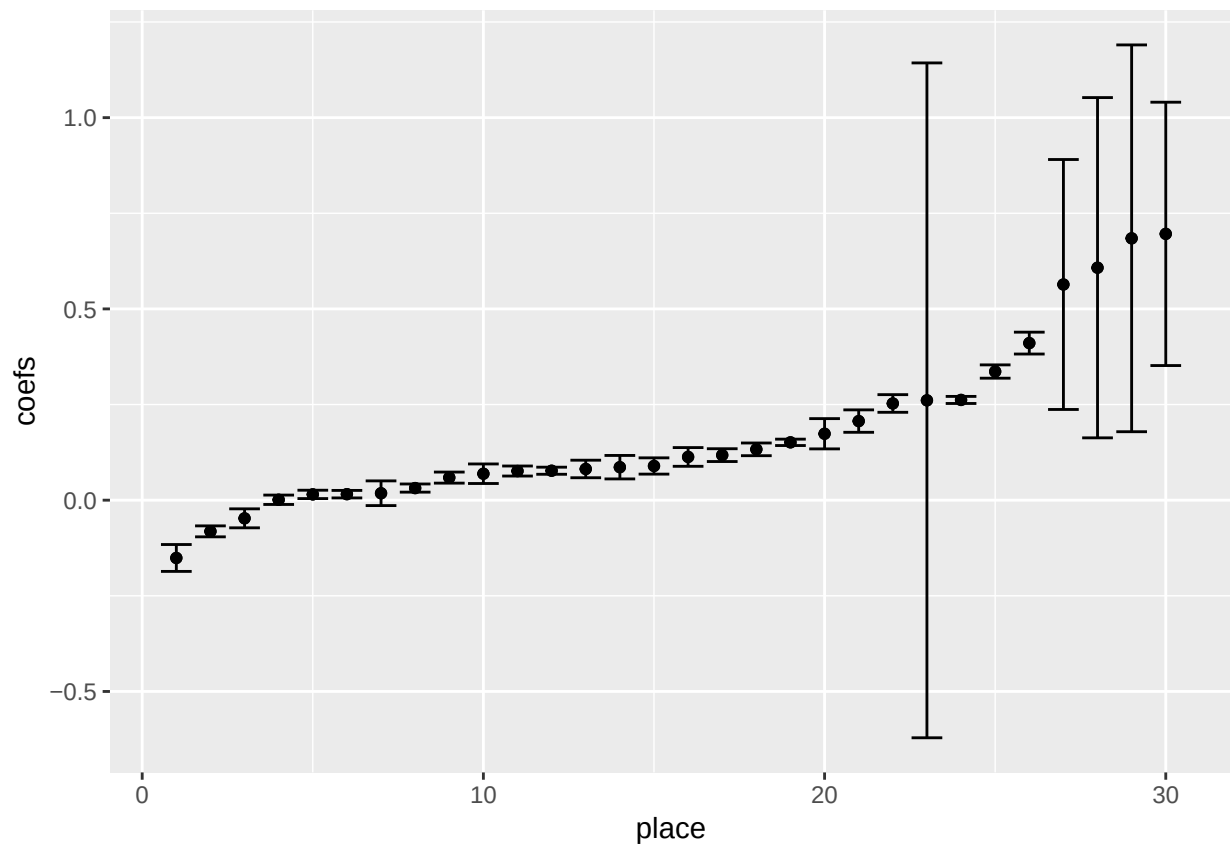
```



```

qplot(place, coefs, data=m)+
  geom_point()+
  geom_errorbar(aes(ymin=coefs-std.err, ymax=coefs+std.err))

```



```
Answer <- qplot(place, coefs, data=m)+
  geom_point()+
  geom_errorbar(aes(ymin=coefs-2*std.err, ymax=coefs+2*std.err))

Answer + ggtitle ("Regression") + xlab("Class Vehicle") + ylab("Coefficient")+theme(plot.title=element_
```

